

6	(a) when two (or more) waves meet (at a point) (resultant) displacement is (vector) sum of individual displacements	B1 B1	[2]
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(b)	(i) $\lambda = ax / D$ (if no formula given and substitution is incorrect then 0/3) $590 \times 10^{-9} = (1.4 \times 10^{-3} \times x) / 2.6$ $x = 1.1 \text{ mm}$	C1 C1 A1	[3]
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(ii)	1. 180° (<i>allow π if rad stated</i>)	A1	[1]
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2.	at maximum, amplitude is 3.4 units and at minimum, 0.6 units <i>intensity</i> $\sim$ <i>amplitude</i> ² allow $I \sim a^2$ ratio = $3.4^2 / 0.6^2$ = 32	C1 C1 A1	[3]
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7	(a) (i) path: reasonable curve upwards between plates	B1	
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